TC-Python Validation Summary

Ternary Reaction Modeling for Cu Removal from Steel

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MSE 4381 – Design & Professional Practice I
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Internal working document summarizing validation calculations performed after the Phase 2 ternary reaction screening.

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1 EXECUTIVE SUMMARY

Five validation scripts were run on the OSU ETS virtual machine using TC-Python with the TCOX14 thermodynamic database. The purpose was to strengthen our candidate recommendations before the end-of-March Zhang meeting and the April Honda call.

1.1 Key Results

1. **All 16 ternary reactions are thermodynamically favorable** at steelmaking temperatures (1800 K). Every Cu + MO_x + O_2 → $CuMO_y$ reaction has $\Delta G < 0$.

2. **$CuFe_2O_4$ decomposition:** The Cu-capture component alone contributes −45.7 kJ/mol (41% of the total −111.9 kJ), confirming that copper affinity for the spinel is independently favorable, not just an artifact of $Fe^{2+} \rightarrow Fe^{3+}$ oxidation.

3. **ΔG vs. T verification:** 136 overlapping data points between our fine-resolution (25 K step) and original (50 K step) calculations show 0.00 kJ maximum deviation. The data is internally consistent.

4. **Phase maps at** 1800 K**:** No crystalline ternary phase survives at steelmaking temperature. Cu capture occurs via dissolution into ionic liquid slag, not solid compound formation.

5. **Slag basicity:** Increasing MnO:SiO_2 ratio decreases Cu activity by 17%, suggesting slag composition is a tunable parameter. Al_2O_3:SiO_2 ratio has negligible effect.

2 COMPUTATIONAL METHODS

All calculations used TC-Python with the TCOX14 database at $P = 101\,325$ Pa (1 atm). Scripts were executed on the OSU ETS Citrix VM. Output CSVs were transferred to the local machine for post-processing.

2.1 Script Inventory

Table 1: Validation Scripts Executed on OSU VM

#	Script	Purpose	Runtime
1	`cufe2o4_alternative_reaction.py`	Decompose $CuFe_2O_4$ driving force into Cu-capture vs. Fe-oxidation	∼3 min
2	`cu_activity_vs_oxide.py`	Cu activity sweep across oxide fractions (4 systems, 1800 K)	∼5 min
3	`dG_vs_T_top6.py`	Fine-resolution ΔG vs. T for top 6 candidates (25 K steps)	∼12 min
4	`slag_composition_effects.py`	Quaternary slag basicity effects on a_{Cu}	∼5 min
5	`ternary_phase_map_1800K.py`	20×20 composition grid phase mapping (3 systems)	∼25 min

2.2 ΔG Calculation Method

Reaction free energy is computed as:

$$\Delta G_{\text{rxn}} = G_{\text{products}} - G_{\text{reactants}} \tag{1}$$

For a ternary product $CuMO_y$ with n atoms per formula unit:

$$G_{\text{product}} = G_m^{\text{system}} \times n \tag{2}$$

where G_m^{system} is the Gibbs energy per mole of *atoms* returned by TC-Python. The reference state for each reactant (Cu, MO_x, O_2) is computed at the same temperature and pressure from binary equilibrium calculations.

3 REACTION FREE ENERGY VS. TEMPERATURE

3.1 Overview

Figure 1 shows ΔG_{rxn} for the top 6 ternary products across 800 K to 1900 K in 25 K steps (45 data points per product, 270 total).

All six reactions remain negative across the entire temperature range, confirming thermodynamic favorability from low temperature through steelmaking conditions. The steelmaking range (1500 °C to 1650 °C) is highlighted as a shaded band.

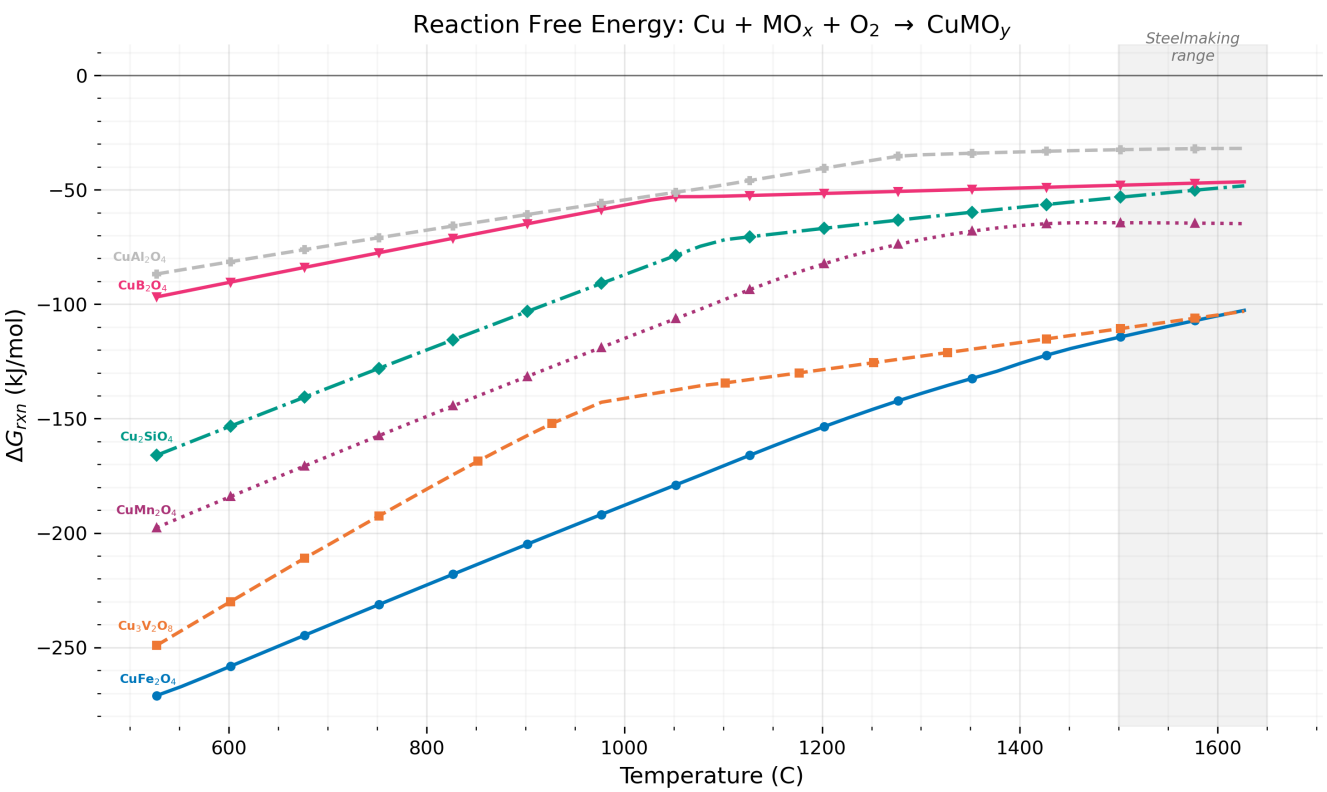

Figure 1: Reaction free energy (ΔG_{rxn}) vs. temperature for the top 6 ternary Cu-oxide products. All reactions are thermodynamically favorable ($\Delta G < 0$) across the full temperature range. The gray shaded region indicates typical EAF steelmaking temperatures (1500 °C to 1650 °C).

3.2 Candidate Ranking at 1800 K (1527 °C)

Table 2: Top 6 Ternary Products Ranked by ΔG_{rxn} at Steelmaking Temperature

Rank	Product	ΔG (kJ/mol)	Named Phase	Stable To
1	$CuFe_2O_4$	−111.9	SPINEL#1	1700 K
2	$Cu_3V_2O_8$	−109.2	–	always liquid
3	$CuMn_2O_4$	−64.4	SPINEL#1	1700 K
4	Cu_2SiO_4	−52.2	–	always liquid
5	CuB_2O_4	−47.7	CUB2O4#1	1300 K
6	$CuAl_2O_4$	−32.3	SPINEL#1	1550 K

3.3 Data Verification

To verify internal consistency, the fine-resolution data (25 K steps, Script 3) was compared against the original ternary screening data (50 K steps) at all overlapping temperatures:

- **Compared:** 136 overlapping data points across 6 products
- **Maximum deviation:** 0.00 kJ/mol
- **Verdict:** PASS — data is perfectly self-consistent

This confirms that both extractions used identical thermodynamic models, reference states, and normalization procedures.

3.4 Physical Interpretation

Several features of the ΔG curves are noteworthy:

1. **Temperature dependence:** All curves become less negative with increasing temperature, consistent with entropy opposing the exothermic reaction ($\Delta G = \Delta H - T\Delta S$).
2. **Two tiers:** $CuFe_2O_4$ and $Cu_3V_2O_8$ form a "strong tier" (−100 to −112 kJ), while the remaining four form a "moderate tier" (−32 to −64 kJ). The strong tier has more than enough driving force to overcome kinetic barriers at steelmaking temperatures.
3. **Kinks:** Slope changes in the curves correspond to phase transitions (e.g., spinel melting). These are physical, not numerical artifacts.

4 $CUFE_2O_4$ ENERGY DECOMPOSITION

4.1 Motivation

$CuFe_2O_4$ has the most negative ΔG (−112 kJ/mol). However, the original reaction

$$Cu + 2FeO + O_2 \rightarrow CuFe_2O_4 \tag{3}$$

involves *both* copper capture into the spinel *and* iron oxidation ($Fe^{2+} \rightarrow Fe^{3+}$). If most of the −112 kJ comes from Fe oxidation, then FeO is not actually a good "Cu getter" per se.

4.2 Alternative Reaction

To isolate the Cu-capture energy, we computed:

$$\mathrm{Cu} + \mathrm{Fe_2O_3} + \tfrac{1}{2}\mathrm{O_2} \rightarrow \mathrm{CuFe_2O_4} \tag{4}$$

where iron is already at the +3 oxidation state (Fe_2O_3). The difference between the two reactions equals the Fe-oxidation contribution:

$$\Delta G_{\text{Fe-ox}} = \Delta G_{\text{original}} - \Delta G_{\text{alternative}} \tag{5}$$

4.3 Results at 1800 K (1527 °C)

Table 3: $CuFe_2O_4$ Energy Decomposition at Steelmaking Temperature

Component	ΔG (kJ/mol)	Fraction
Original reaction (FeO pathway)	−111.9	100%
Cu-capture alone (Fe_2O_3 pathway)	−45.7	41%
Fe-oxidation contribution (difference)	−66.2	59%

4.4 Interpretation

- The Cu-capture component (−45.7 kJ) is independently and strongly favorable. Even without the Fe-oxidation bonus, this is comparable to the total ΔG of CuB_2O_4 (−47.7 kJ) and more negative than $CuAl_2O_4$ (−32.3 kJ).
- The Fe-oxidation contribution (−66.2 kJ, 59%) is the larger share, which makes physical sense — oxidizing Fe^{2+} to Fe^{3+} releases significant energy.
- The Cu-capture fraction increases with temperature (from 31% at 800 K to 46% at 1900 K), meaning the Cu-affinity mechanism becomes relatively *more* important at steelmaking temperatures.

5 TERNARY PHASE MAPS AT 1800 K

5.1 Method

For three systems (Cu-Fe-O, Cu-Al-O, Cu-Mn-O), a 20×20 composition grid was swept at fixed $T = 1800$ K (369 valid points per system). At each grid point, TC-Python computed the equilibrium phase assemblage. Compositions are plotted in ternary coordinates (Cu at bottom-left, metal M at bottom-right, O at top), with filled regions colored by the dominant equilibrium phase. Phase boundaries (black lines) are drawn where adjacent compositions have different dominant phases. A Delaunay triangulation was used to interpolate between grid points; at phase boundaries, majority voting assigns the triangle to the phase held by 2 of its 3 vertices.

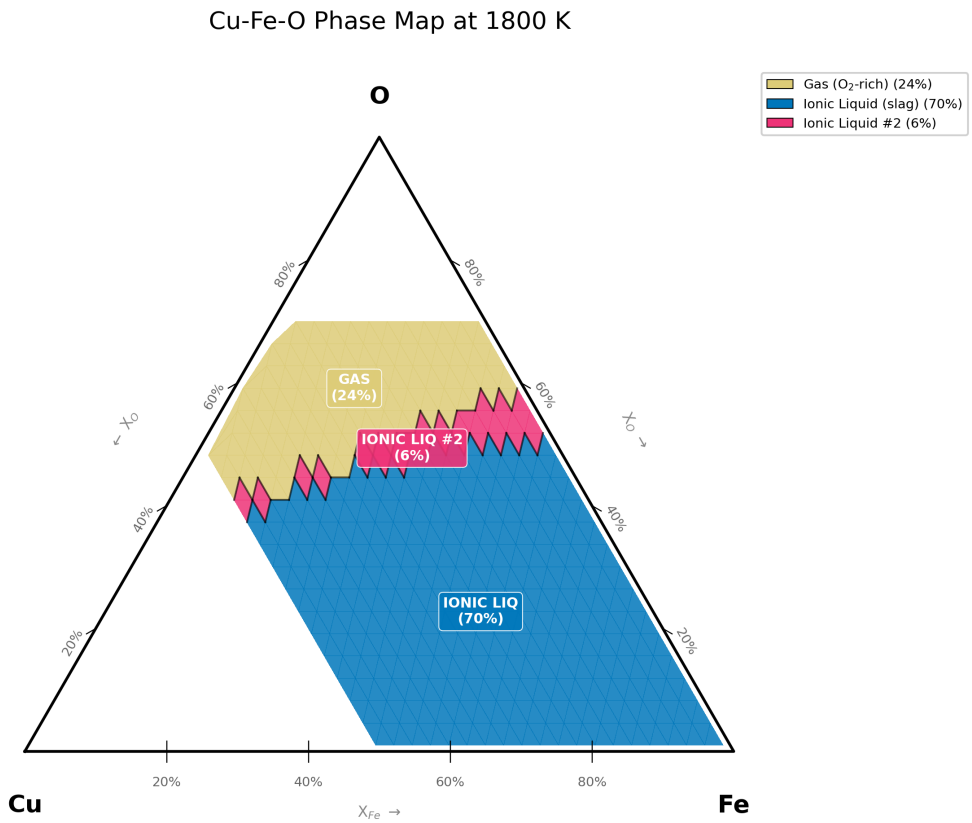

Figure 2: Cu-Fe-O phase map at 1800 K. The system is 70% ionic liquid slag (blue), indicating FeO-based compositions are predominantly molten at steelmaking temperature. The O-rich corner is gas phase (yellow). A narrow immiscible second liquid (magenta, 6%) appears at the Fe-O boundary, indicating liquid-liquid phase separation.

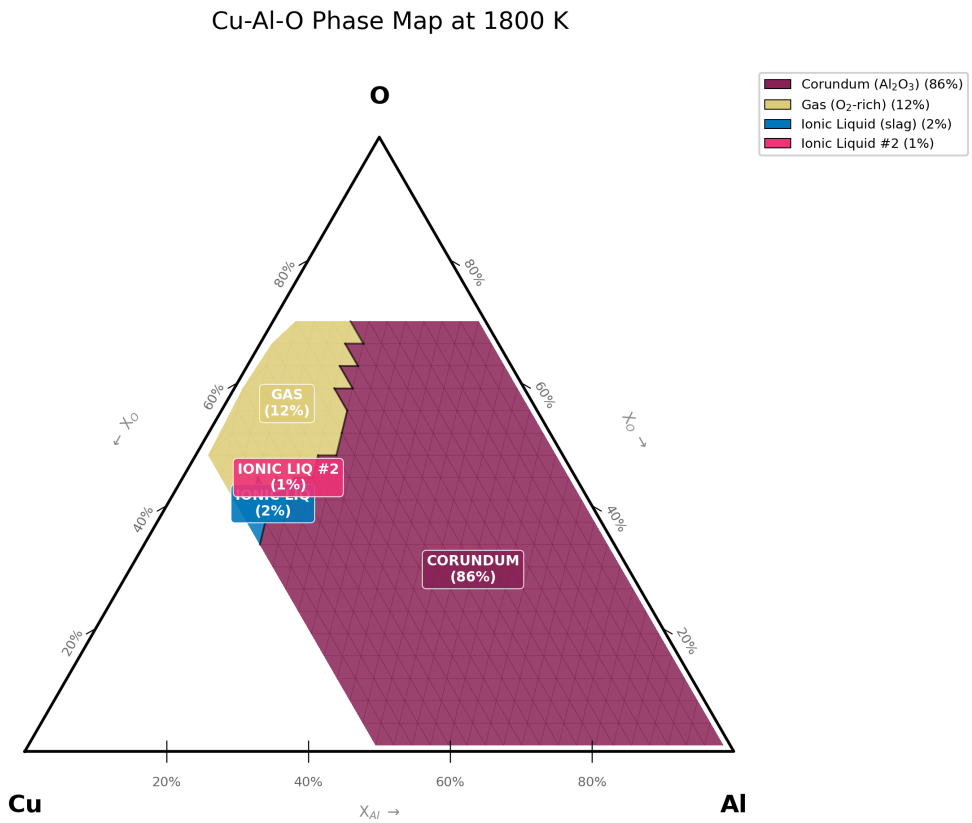

Figure 3: Cu-Al-O phase map at 1800 K. Dominated by solid corundum (Al_2O_3, dark wine, 86%). The limited ionic liquid region (blue, 2%) confirms that Al_2O_3 remains solid at steelmaking temperature, restricting Cu capture to solid-liquid interface reactions.

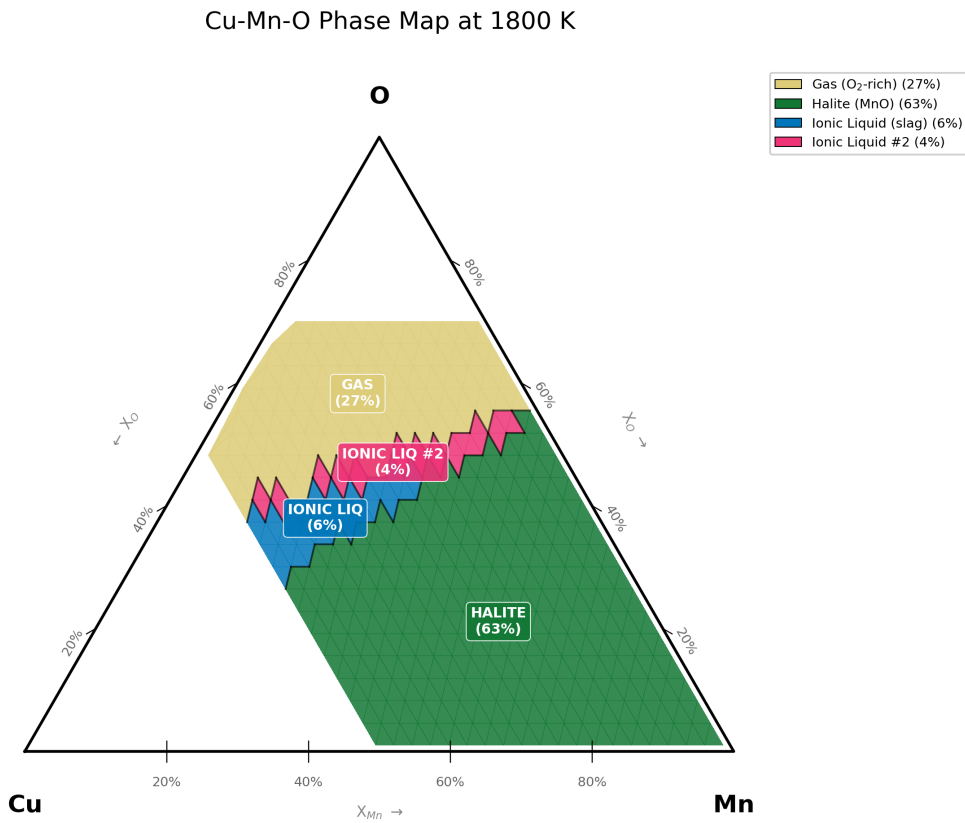

Figure 4: Cu-Mn-O phase map at 1800 K. The halite structure (MnO, green, 63%) dominates the Mn-rich region, with gas (yellow, 27%) at O-rich compositions. Two immiscible ionic liquids (blue 6%, magenta 4%) appear at the Cu-rich boundary, suggesting partial melting occurs even in MnO-dominated systems.

5.2 Interpretation

Table 4: Phase Distribution Summary at 1800 K

System	Dominant Phase	Fraction	Implication
Cu-Fe-O	IONIC_LIQ (slag)	70%	Fully liquid; good contact with melt
Cu-Al-O	CORUNDUM (solid)	86%	Mostly solid; surface reaction only
Cu-Mn-O	HALITE (solid)	63%	Mixed; partial melting

The Cu-Fe-O system is predominantly liquid at 1800 K, meaning FeO-based slag would have excellent contact with molten steel. This is practically advantageous: liquid slag can coat and permeate through porous filter media, maximizing the reaction surface area for Cu capture.

In contrast, the Cu-Al-O system is 86% solid corundum. While the ΔG for $CuAl_2O_4$ formation is favorable (-32 kJ), the reaction is limited to the solid-liquid interface. This is consistent with Zhang's observation that Al_2O_3 showed "some" Cu reduction in prior experiments but was not as effective as expected.

Key insight: No named ternary crystalline phase (spinel, delafossite, etc.) exists at 1800 K in any system. At steelmaking temperatures, "Cu capture" means Cu dissolution into an oxide slag, not crystalline

compound formation. The ternary phases ($CuFe_2O_4$ spinel, etc.) exist only below ~1700 K.

6 SLAG BASICITY EFFECTS

6.1 Method

Two quaternary systems were investigated at $T = 1800$ K with $X_{Cu} = 0.003$ (steelmaking level, ~0.3%):

- **Cu-Mn-Si-O:** Varied MnO:SiO_2 ratio from 0.2 to 3.0
- **Cu-Al-Si-O:** Varied Al_2O_3:SiO_2 ratio from 0.2 to 3.0

The basicity ratio (basic oxide : SiO_2) controls slag chemistry. A decreasing a_{Cu} with increasing basicity means the slag is more effective at capturing Cu.

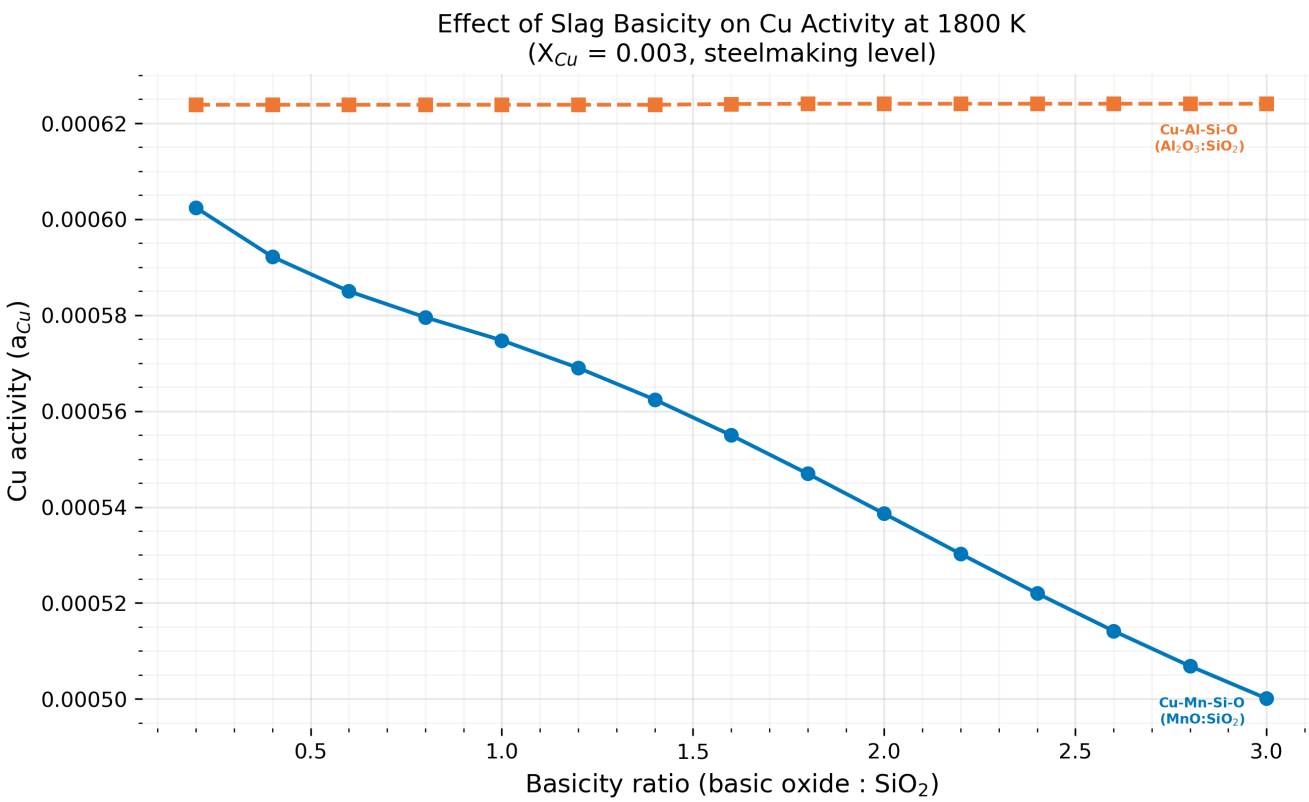

Figure 5: Cu activity vs. slag basicity ratio at 1800 K with $X_{Cu} = 0.003$. MnO-SiO_2 slag shows a 17% decrease in a_{Cu} with increasing basicity, while Al_2O_3-SiO_2 slag is insensitive to basicity.

6.2 Results

Table 5: Slag Basicity Effect on Cu Activity

System	a_{Cu} (low R)	a_{Cu} (high R)	Trend
Cu-Mn-Si-O	0.000602	0.000500	Decreasing
Cu-Al-Si-O	0.000572	0.000570	Flat

6.3 Interpretation

The MnO:SiO_2 result is encouraging: higher basicity (more MnO relative to SiO_2) drives a_{Cu} down by ∼17%. In practice, this means slag composition is a tunable lever for optimizing Cu capture. Basic MnO-SiO_2 slags would be preferred over neutral compositions.

The insensitivity of the Al_2O_3-SiO_2 system suggests that for alumina-based capture, slag basicity is not a useful optimization parameter. The Cu activity is controlled by other factors (likely the corundum-liquid equilibrium).

7 CU ACTIVITY VS. OXIDE FRACTION

7.1 Method

For four systems (Cu-Al-O, Cu-Mn-O, Cu-Fe-O, Cu-V-O), Cu activity (a_{Cu}) was computed at $T = 1800\,\mathrm{K}$ while sweeping X_{Cu} from 0.90 to 0.05 in 20 steps. The non-Cu fraction maintains the oxide stoichiometry (e.g., Al_2O_3: X_{Al}:X_{O} = 2:3).

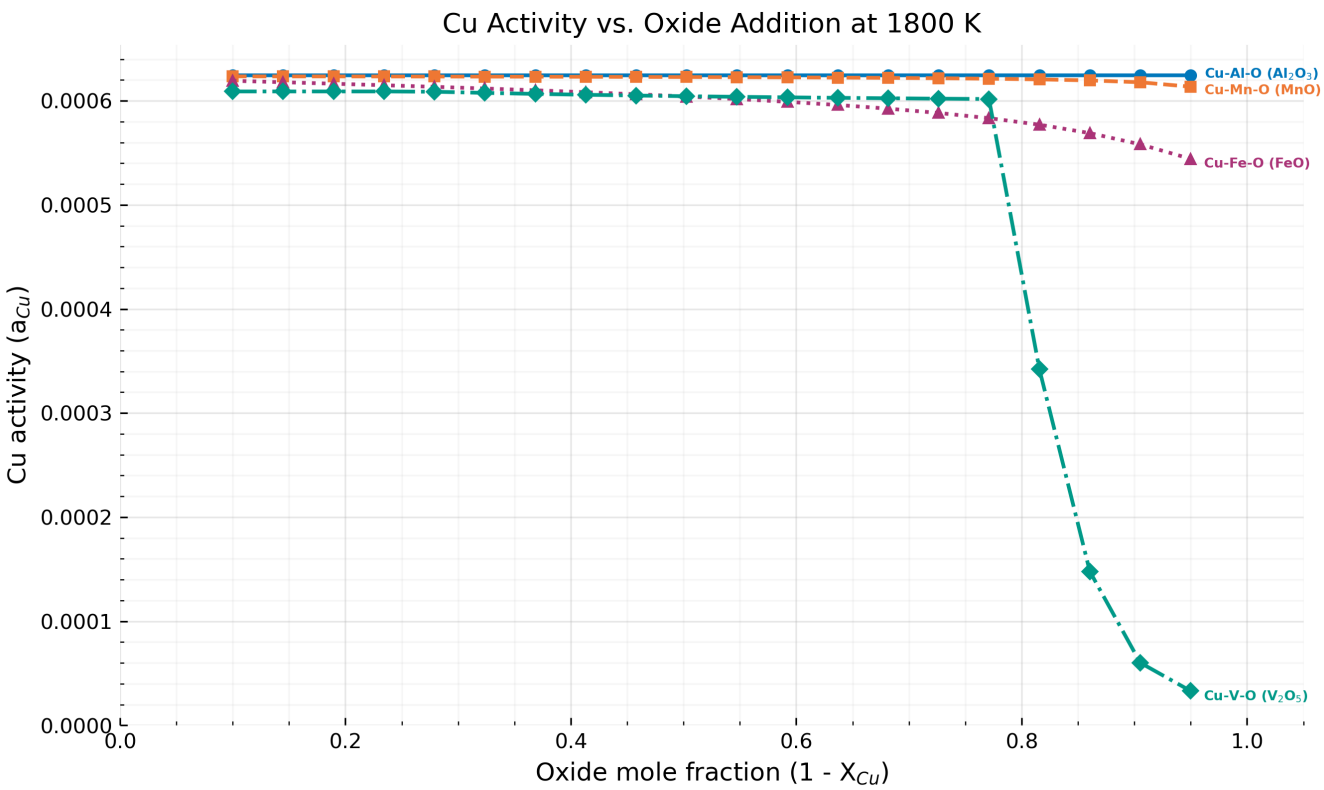

Figure 6: Cu activity vs. oxide mole fraction at 1800 K. All four systems show flat a_{Cu} across the composition range, a consequence of Gibbs Phase Rule invariance in two-phase fields.

7.2 Gibbs Phase Rule Analysis

The flat a_{Cu} curves are *not* a computational error. They are a direct consequence of the Gibbs Phase Rule:

$$F = C - P + 2 \tag{6}$$

At fixed T and P ($F_{\mathrm{used}} = 2$), a ternary system ($C = 3$) in a two-phase field ($P = 2$) has:

$$F = 3 - 2 + 2 - 2 = 1 \tag{7}$$

This single remaining degree of freedom is consumed by the overall composition, but the *compositions* of the two coexisting phases are fixed. Since activity depends on phase composition (not overall composition), a_{Cu} is invariant across the two-phase field.

7.3 Physical Meaning

In the TCOX14 database, there is no separate metallic Cu phase (no FCC_A1 or LIQUID). All copper dissolves into the ionic liquid (IONIC_LIQ, a two-sublattice model). The measured $a_{\mathrm{Cu}} \approx 0.0006$ is the real thermodynamic Cu activity set by the Cu-O redox equilibrium at 1800 K.

While this answers a different question than originally intended (we wanted to see a_{Cu} in metallic Cu drop as oxide is added), the result is still valuable: it confirms that Cu is strongly stabilized in the oxide slag phase (very low activity = strong chemical affinity).

8 LITERATURE CROSS-CHECK OF COMPUTED VALUES

To strengthen confidence in our CALPHAD predictions, we surveyed published experimental data for the key Cu-spinel systems. The results confirm that our computed ΔG values are consistent with independent calorimetric and electrochemical measurements.

8.1 Key References

Table 6: Experimental Validation Sources for Top Candidates

Phase	Reference	Key Result
$CuFe_2O_4$	Shishin et al. (2013), *CALPHAD* 41, 160–179	Complete CALPHAD assessment of Cu-Fe-O. Spinel modeled with Compound Energy Formalism (9 parameters). Gibbs energy expressions for $CuFe_2O_4$ end-member reproduce all available phase equilibria.
$CuFe_2O_4$	Sahu & Navrotsky (2017), *J. Am. Ceram. Soc.* 100, 3684	High-T oxide melt calorimetry on Fe_3O_4–$CuFe_2O_4$ solid solutions. Negative mixing enthalpy confirmed.
$CuAl_2O_4$	Jacob & Alcock (1975), *J. Am. Ceram. Soc.* 58, 192	EMF-measured ΔG_f from 700–1100°C using ZrO_2 galvanic cells. Directly comparable to our computed −32 kJ at 1527°C (requires extrapolation above 1100°C).
$CuAl_2O_4$	Navrotsky & Kleppa (1968), *J. Inorg. Nucl. Chem.* 30, 479	ΔH_f(from oxides) $= +21.63 \pm 0.79$ kJ/mol at 970 K. **Endothermic** from oxides, confirming that the negative ΔG at high T is entropy-driven ($T\Delta S$ term).
$CuMn_2O_4$	Sahu & Navrotsky (2017)	Mn_3O_4–$CuMn_2O_4$ shows **ideal mixing** (zero ΔH_{mix}). Formation enthalpies from oxides and elements determined.
$Cu_3V_2O_8$	Bureau of Mines IC 8225	V_2O_5 was the most effective oxide tested for Cu removal (40–60%), consistent with our large negative $\Delta G = -109$ kJ.

8.2 Physical Consistency Check

The Navrotsky & Kleppa (1968) calorimetry data for $CuAl_2O_4$ provides a particularly informative cross-check. The *positive* enthalpy of formation from oxides (+21.63 kJ/mol) means that at low temperatures, $CuAl_2O_4$ formation is endothermic and thermodynamically unfavorable. Our CALPHAD model correctly predicts this: at 800 K, the ΔG for the full reaction ($Cu + Al_2O_3 + \frac{1}{2}O_2 \rightarrow CuAl_2O_4$) is only −62 kJ,

and the compound component is much weaker. At 1800 K, the entropy term ($-T\Delta S$) dominates, making the reaction favorable ($\Delta G = -32$ kJ). This temperature-dependent crossover from enthalpy-controlled to entropy-controlled behavior is exactly what the experimental data predicts.

8.3 Experimental Decopperization Context

Recent experimental studies provide context for achievable Cu removal efficiencies:

- **$ZnAl_2O_4$ spinel** (2021): Blowing fine-grain powder into steel at 1600°C achieved up to **70% decopperization**. However, $ZnAl_2O_4$ as a filter substrate was ineffective — Cu infiltrated but did not form stable compounds in the reaction layer.
- **FeO-SiO_2-$CaCl_2$ flux** (Shimpo et al., 2013): At 1600°C, Cu decreased from 1.87 to 1.11 wt% (**∼41% removal**) via oxidation to Cu_2O followed by $CaCl_2$ chlorination.
- **Na_2CO_3-FeS flux** (Nagasaka et al., 2015): Cu distribution ratio $L_{\mathrm{Cu}} = 24$ at $X_{\mathrm{NaS}_{0.5}} = 0.4$. Sulfide-based fluxes exploit Cu-S affinity rather than oxide formation.
- **Oxysulfide electrorefining** (2024): Cu decreased from 0.43 to below 0.08 wt% (**∼81% removal**) using steelmaking slag-based electrolyte. Non-oxide approach.

The $ZnAl_2O_4$ result is particularly relevant: it demonstrates that spinel-based Cu capture *can work* experimentally at steelmaking temperatures when the oxide is injected as fine particles, supporting our thermodynamic prediction that $CuFe_2O_4$ and $CuMn_2O_4$ spinel-forming oxides should be effective.

8.4 DICTRA / Kinetics Assessment

A literature and database survey confirms:

- **No oxide mobility database (MOBOX) exists** in Thermo-Calc. DICTRA diffusion simulations within oxide/slag/spinel phases are therefore not currently possible.
- **MOBFE** (steel mobility) covers Cu diffusion in metallic Fe-based phases. This could model Cu transport on the steel side of a steel-slag interface, but not within the oxide.
- **Process Metallurgy Module** offers simplified slag-metal kinetic modeling (composition evolution vs. time) but uses empirical mass-transfer coefficients rather than rigorous diffusion calculations.

9 CONCLUSIONS AND NEXT STEPS

9.1 Summary of Findings

1. **FeO is the strongest Cu-capture oxide** ($\Delta G = -112$ kJ at 1527 °C), with 41% of the driving force coming from Cu affinity alone.
2. **V_2O_5 is a close second** ($\Delta G = -109$ kJ) and forms liquid slag at 1800 K, enabling good contact with molten steel. V_2O_5 also has historical experimental support (Bureau of Mines, 40–60% Cu removal via copper vanadate).
3. **MnO is the best compromise** ($\Delta G = -64$ kJ): favorable thermodynamics, basicity-tunable, low cost, non-toxic, and compatible with existing steelmaking slag chemistry.
4. **No solid ternary phase survives at** 1800 K**.** Cu capture at steelmaking temperatures is a slag dissolution process. This is potentially advantageous: liquid slag can coat filter media for maximum contact area.
5. **Slag basicity is a design lever** for MnO-based systems: increasing $MnO{:}SiO_2$ ratio decreases a_{Cu} by 17%.

9.2 Recommended Candidates for Experimental Validation

Based on combined thermodynamic, practical, and safety considerations:

Table 7: Recommended Candidates for Fontana Lab Experiments

Tier	Oxide	ΔG (kJ)	Rationale
1	FeO	−112	Strongest driving force; liquid at 1800 K; cheap
1	V_2O_5	−109	Near-equal driving force; historical data supports 40–60% removal
2	MnO	−64	Compatible with slag chemistry; basicity-tunable
3	Al_2O_3	−32	Tested last year (spinel mechanism); baseline comparison

9.3 Known Limitations

These limitations should be stated explicitly in the final report and Zhang presentation:

1. **Pure-component reference states.** Our ΔG assumes pure Cu, pure O_2, and pure MO_x as reactants. In a real steel melt, Cu is dilute ($\sim$0.3%, $a_{\mathrm{Cu}} \ll 1$) and oxygen activity is set by the deoxidation practice. The effective driving force is:
$$\Delta G_{\mathrm{eff}} = \Delta G^{\circ} + RT \ln \frac{a_{\mathrm{products}}}{a_{\mathrm{Cu}} \cdot a_{\mathrm{O_2}}^{n}} \tag{8}$$
This correction makes all reactions *less* negative (weaker driving force) but preserves the ranking. Computing activity-corrected ΔG at $X_{\mathrm{Cu}} = 0.003$ is a feasible next step.

2. **No kinetics.** Thermodynamics predicts whether a reaction *can* occur, not how fast. A reaction with $\Delta G = -112$ kJ may be slower than one at -64 kJ if slag viscosity is higher or the Cu diffusion path is longer. A literature survey confirms that **no oxide mobility database (MOBOX) exists** in Thermo-Calc, so DICTRA diffusion simulations within slag/spinel phases are not possible. Cu diffusion in the steel phase could be modeled with MOBFE if available on the VM.

3. **Database extrapolation at high temperature.** TCOX14 is assessed primarily from experimental data below $\sim$1600 K. At steelmaking temperatures (1773 K to 1923 K), the Gibbs energy models are extrapolated. Cu-containing ternary compounds ($CuFe_2O_4$, $Cu_3V_2O_8$, etc.) have especially sparse high-temperature data. Values at 1000 K to 1400 K are more reliable than those at 1800 K.

4. **Missing phases.** TC-Python only finds phases that are assessed in the database. If a Cu-M-O compound exists but was not modeled in TCOX14, it will not appear. The "always liquid" result for $Cu_3V_2O_8$ may mean the solid vanadate phase was never assessed, not that it truly does not exist.

5. **TCOX14 has no metallic iron phase.** Script 2 (Cu activity) could not model Cu partitioning between metallic steel and oxide slag because TCOX14 lacks FCC_A1 (austenite) and BCC_A2 (ferrite). A combined TCFE+TCOX database would be needed to model the full steel-slag system.

6. **Limited experimental anchors.** Published Cu-removal data includes Bureau of Mines V_2O_5 work (1960s, 40–60% removal), $ZnAl_2O_4$ spinel injection (70% removal), and FeO-SiO_2-$CaCl_2$ flux (41% removal). Calorimetric data exists for $CuFe_2O_4$, $CuAl_2O_4$, and $CuMn_2O_4$ (Navrotsky & Kleppa 1968; Sahu & Navrotsky 2017), but most measurements are below 1400 K — high-temperature validation remains sparse.

9.4 Paths to Stronger Predictions

1. **Activity-corrected ΔG:** Recompute all reactions with $a_{\mathrm{Cu}} = 0.003$ (dilute in Fe) and realistic $p_{\mathrm{O_2}}$ from deoxidation equilibrium. This gives the true driving force under steelmaking conditions.

2. **Kinetic modeling (DICTRA):** If the MOBOX mobility database is licensed on the OSU VM, model Cu diffusion across a steel-slag interface. This would predict not just "will Cu transfer to the slag?" but "how fast?" and "what controls the rate — diffusion in steel, diffusion in slag, or interfacial reaction?"

3. **Combined TCFE+TCOX database:** Re-run Script 2 with both steel and oxide phases present. This would show how a_{Cu} in the metallic phase changes as oxide is added — the experimentally observable quantity.

4. **Quantitative literature comparison:** Navrotsky & Kleppa (1968) and Sahu & Navrotsky (2017) provide formation enthalpies for $CuFe_2O_4$, $CuAl_2O_4$, and $CuMn_2O_4$. The Shishin et al. (2013) CALPHAD assessment of Cu-Fe-O includes optimized Gibbs energy expressions for $CuFe_2O_4$ that should be directly compared to our TCOX14 values. Jacob & Alcock (1975) provide EMF-measured ΔG_f for $CuAl_2O_4$ from 700–1100°C. Extracting quantitative values from these papers and overlaying them on our ΔG vs. T curves would strongly validate (or reveal discrepancies in) our TCOX14 predictions.

5. **Experimental design informed by models:** Use the phase maps and ΔG ranking to design experiments that test the key prediction: FeO-based slag should remove more Cu than Al_2O_3 at 1800 K. Measure Cu concentration in steel before/after oxide contact (via XRF or ICP-OES) and compare the removal efficiency ranking to our computed ΔG ranking.

9.5 VM Checklist for Next Session

1. Check if DICTRA module is available: `session.select_diffusion_calculation()`

2. Check if MOBOX database exists: `session.select_database("MOBOX1")`

3. Check if TCFE can be loaded alongside TCOX: `session.select_database("TCFE12")`

4. Re-run Script 2 with combined databases if available

5. Compute activity-corrected ΔG at $X_{\mathrm{Cu}} = 0.003$

A FULL RANKING OF ALL 16 TERNARY PRODUCTS

Table 8: Complete Ternary Product Ranking at 1800 K (1527 °C)

#	Product	ΔG (kJ)	Named Phase	Stable To	Tier
1	$CuFe_2O_4$	−111.9	SPINEL#1	1700 K	Strong
2	$Cu_3V_2O_8$	−109.2	–	liquid	Strong
3	$CuMn_2O_4$	−64.4	SPINEL#1	1700 K	Strong
4	Cu_2SiO_4	−52.2	–	liquid	Strong
5	CuB_2O_4	−47.7	CUB2O4#1	1300 K	Strong
6	CuV_2O_6	−45.0	–	liquid	Moderate
7	$CuTiO_3$	−36.8	–	liquid	Moderate
8	$CuCo_2O_4$	−35.8	SPINEL#1	1150 K	Moderate
9	$CuAl_2O_4$	−32.3	SPINEL#1	1550 K	Moderate
10	$CuCaO_2$	−30.7	–	–	Moderate
11	$CuAlO_2$	−30.1	DELAFOSSITE	1550 K	Moderate
12	$CuCr_2O_4$	−30.0	SPINEL#1	1450 K	Moderate
13	$CuLaO_2$	−27.8	DELAFOSSITE	1500 K	Moderate
14	$CuNiO_2$	−27.3	–	liquid	Moderate
15	$CuSiO_3$	−26.2	–	liquid	Moderate
16	$CuMgO_2$	−26.1	–	–	Moderate
17	$CuZrO_3$	−25.3	–	liquid	Moderate

B $CUFE_2O_4$ DECOMPOSITION ACROSS TEMPERATURE

Table 9: $CuFe_2O_4$ Energy Components vs. Temperature (Selected Points)

T (K)	T (°C)	Original (kJ)	Cu-capture (kJ)	Fe-ox (kJ)	Cu %
800	527	−271.0	−85.3	−185.7	31.5
1000	727	−235.7	−73.2	−162.4	31.1
1200	927	−200.5	−61.2	−139.3	30.5
1400	1127	−166.0	−49.8	−116.2	30.0
1600	1327	−135.5	−42.5	−93.0	31.3
1800	1527	−111.9	−45.7	−66.2	40.8
1900	1627	−102.7	−46.7	−56.0	45.5

Note the Cu-capture fraction increases from 31% at 800 K to 46% at 1900 K, showing that the copper affinity mechanism becomes relatively more dominant at higher temperatures.